

ACADEMIC PROJECT • 2026

Food Waste Analysis and Economic Loss Prediction

Data Analytics & Machine Learning Project

NTI X ITIDA Advanced Data Analytics Internship

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Project Overview

Every year, countries produce huge amounts of food — and a large share of it ends up as food waste.

01

Looks environmental

At first glance, food waste seems like a purely environmental issue.

02

It's economic too

Every wasted ton once had real value — so waste means hidden financial loss.

03

Data can connect them

Waste patterns and economic loss are linked, and that link can be measured.

So the question becomes:

Can data reveal where and why food waste happens, link it to economic impact, and predict the cases where the loss will be highest?

THIS PROJECT

Food Waste Analysis and Economic Loss Prediction

Dataset

Country GEOGRAPHY	Year TIME	Food Category CATEGORY
Total Waste (Tons) VOLUME	Avg Waste per Capita (Kg) INTENSITY	Population (Million) SCALE
Household Waste (%) SHARE		

PREDICTION TARGET

Economic
Loss

Million \$

What financial impact should we expect?

Exploratory Data Analysis

5

columns contained missing values

- ✓ No duplicate rows
- ✓ No invalid negative values
- ✓ No IQR outliers detected



PROFILE → VALIDATE → VISUALIZE → INTERPRET

Data Cleaning

Population		Median by Country + Year
Total Waste		Median by Country + Food Category
Avg Waste per Capita		Median by Country + Food Category
Household Waste		Median by Country
Food Category		Mode
FALLBACK • GLOBAL MEDIAN		

Missing Values: Before vs After

BEFORE CLEANING

Column	Missing Count	Missing %
Population (Million)	600	12.0%
Household Waste (%)	500	10.0%
Total Waste (Tons)	400	8.0%
Food Category	300	6.0%
Avg Waste per Capita (Kg)	250	5.0%

Every gap was closed with group-based imputation, not simple deletion.

AFTER CLEANING

0

missing values across every column

- ✓ Country
- ✓ Year
- ✓ Food Category
- ✓ Total Waste (Tons)
- ✓ Economic Loss (Million \$)
- ✓ Avg Waste per Capita (Kg)
- ✓ Population (Million)
- ✓ Household Waste (%)

Leakage-Safe ML Preparation



“Train data was used to calculate imputation values to prevent Data Leakage.”

Power BI Dashboard

From analysis to action.

Food Waste Analysis Dashboard

Common Features

- KPIs: Total Population | Total Economic Loss (\$) | Average Individual Food Waste (kg) | Total Food Waste (Tons)
- Slicers: Year | Country | Food Category
- Navigation: Waste Overview | Country Analysis | Map

Page 1 — Waste Overview

- Household Food Waste (%) — Gauge
- Economic Loss (\$) by Year — Area Line Chart
- Food Waste (Tons) by Year — Area Line Chart
- Food Waste (Tons) by Category — Funnel Chart

Page 2 — Country Analysis

- Top 10 Countries by Food Waste (Tons)
- Top 10 Countries by Economic Loss (\$)
- Top 10 Countries by Household Food Waste (%)
- Top 10 Countries by Individual Food Waste (Kg)

Page 3 — Map

- Economic Loss by Country — Map

Power BI Dashboard



Power BI Dashboard



Power BI Dashboard



Linear Regression

How much economic loss is expected?

Predicts the actual numerical value of Economic Loss (Million \$).

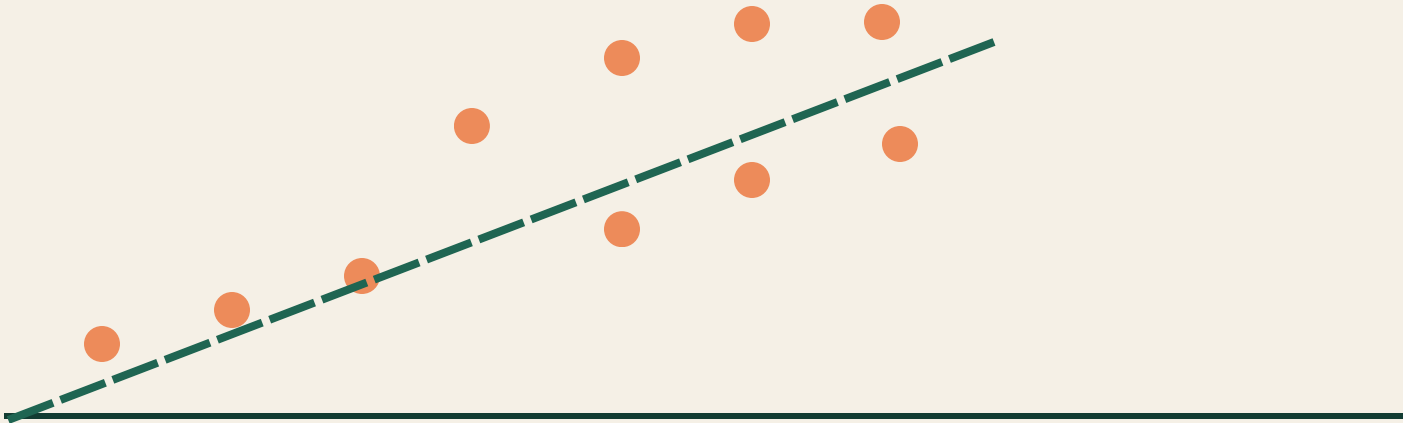
53.32%

R² • explained variance

MAE	6,734.72
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MSE	106,144,200.55
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RMSE	10,302.63
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Logistic Regression

INPUT FEATURES

Waste • population • category

LOGISTIC MODEL

Estimate class probability

PROBABILITY

0.00 → 1.00

CLASS

0 Low / 1 High

Classification answers which level — not the exact loss amount.

Creating the Classification Target

No binary target existed in the original dataset — so loss level was engineered from the training median.

`y_train.median()`

0 LOW LOSS

<

1 HIGH LOSS

>=

TRAINING MEDIAN

Median calculated on training data only → no leakage

Model Comparison

LINEAR REGRESSION

How much?

Predict exact Economic Loss

Continuous numerical value

LOGISTIC REGRESSION

Which level?

Classify Economic Loss level

0 = Low / 1 = High

Model Comparison

LINEAR REGRESSION

```
MAE: 6734.715098657865
MSE: 106144200.55282654
RMSE: 10302.630758831772
R2: 0.5331983305434327
```

LOGISTIC REGRESSION

	precision	recall	f1-score	support
0	0.76	0.81	0.78	493
1	0.80	0.74	0.77	507
accuracy			0.78	1000
macro avg	0.78	0.78	0.78	1000
weighted avg	0.78	0.78	0.78	1000

Links

[Link Deployment](#)

[Link Image](#)
[Project in Docker](#)
[Hub](#)



Conclusion

01	UNDERSTAND	EDA exposed quality issues and meaningful patterns.
02	PRESERVE	Group-based imputation retained data relationships.
03	PROTECT	Training-only preprocessing prevented leakage.
04	PREDICT	Regression estimated loss; classification assigned level.
05	ACT	Power BI turned analysis into interactive intelligence.

Data-driven decisions can support more sustainable food systems.

QUESTIONS & DISCUSSION

Thank You

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